



Politecnico di Milano

Graduate School of Management

New Technologies for Finance — May 2025

**Early-Warning System for Risk-Off Regimes
with Domain-Specific Allocation Routing**

Unsupervised Anomaly Detection for Regime-Aware Portfolio Allocation

Edoardo Ercelli · Luca Galli
Francesco Impenati · Francesco Mazzini

Motivation and Objective

Traditional early-warning systems treat risk-off as a binary decision—equity or cash—ignoring that crises are heterogeneous. In 2008, the USD strengthened (global dollar funding squeeze; Maggiori, 2020); during inflationary stress, gold outperforms as real rates fall (Baur & Lucey, 2010); in early COVID-19, the dollar rallied then reversed as the Fed intervened.

We build an EWS answering *when* to exit risk assets (ensemble anomaly detection) and *where* to allocate (macro-based routing: Cash USD, Gold, or MBS). The strategy deploys **1.5× equity leverage** in calm periods and relies on early detection to rotate out before drawdowns materialize, maximizing the Sharpe ratio. The leverage is justified only if recall is adequate—missing a crisis at 1.5× is substantially costlier than at 1.0×.

Data and Feature Engineering

Bloomberg weekly data, January 2000–April 2021: 1,111 observations, 43 features (7 MSCI equity, 3 FX, 4 commodities, 8 bond TR indices, 18 yields, VIX, ECSURPUS). Target $Y=1$: risk-off (21.3% prevalence), clustered in 2000–02, 2008–09, 2011–12, 2020. No missing values. Proxies: MSCI World→MXUS; Global Agg→LUACTRUU; credit spreads as log price-index ratios.

Stationarity: log-returns for prices, first differences for yields, level for VIX/ECSURPUS; 42/42 pass ADF at 5%. We construct 20 cross-asset spreads (term, BTP-Bund, HY-IG, EM credit, VRP, equity-bond rotation, gold-oil, JPY strength) and remove 7 collinear features ($|\rho|>0.9$), yielding **56 clean features**. Additionally, 14 routing triggers are built across three macroeconomic domains.

Walk-Forward Validation

Purged expanding walk-forward CV with **4-week embargo** (drops 4 observations between train and val, eliminating leakage from Δ_{4w} lookbacks). Development ≤ 2018 ; test holdout 2019–2021 sealed (121 weeks, contains COVID). Five folds: GFC 2008 (53 positives), Euro Debt 2011 (64), Taper 2013 (2), China-Oil 2015–16 (10), Q4 2018 (6). Folds 3–5 are thin because 2013–2018 is a low-volatility regime—a data property. All metrics **weighted by** n_{pos} . Scaler fit only on fold train; AE early-stopping on temporal train sub-split only.

Anomaly Detection Models

Four novelty detectors trained on $Y=0$ weeks only: **(1) MVG**—Mahalanobis distance with Ledoit-Wolf shrinkage (mandatory: 56 features, ~ 360 normals); **(2) One-Class SVM**—RBF kernel, grid $\nu \times \gamma$, best $\nu=0.05$, $\gamma=0.001$; **(3) Autoencoder**—56→24→12→6→12→24→56, dropout 0.15, early stopping on train sub-split; **(4) Isolation Forest**—200 trees, contamination=0.10. Scores converted to empirical percentiles against train-normals for comparability, combined via soft mean voting.

Table 1: Test holdout performance (sealed 2019–2021).

	F1	AUC-ROC	AUC-PR	Prec.	Recall	F2
MVG	0.516	0.886	0.770	1.000	0.348	0.400
One-Class SVM	0.615	0.851	0.732	0.750	0.522	0.556
Isolation Forest	0.583	0.810	0.680	0.560	0.609	0.598
Autoencoder	0.516	0.873	0.764	1.000	0.348	0.400
Soft Mean Ensemble	0.650	0.859	0.742	0.765	0.565	0.596

The ensemble (F1=0.650) outperforms the best single model (SVM, 0.615). Mean pairwise error correlation=0.66, confirming diverse error patterns and genuine ensemble value.

Domain-Specific Routing

When risk-off is signaled, three domain sub-scores determine allocation. The **USD sub-score** (mean of signed z-scores: funding stress +1, DXY +1, VRP +1, Treasury flight −1, US relative +1) captures dollar-as-safe-haven dynamics during global, non-US stress (Avdjiev et al., 2019). The **Gold sub-score** (real yield −1, DXY −1, JPY +1, equity-bond ρ +1, gold-oil +1) activates when stress is monetary/inflationary or the dollar weakens (Baur & Lucey, 2010). Note: DXY appears in both domains with *opposite sign*—same indicator, opposite economic reading. The **MBS sub-score** is rule-based: active only in moderate stress ($VIX \in [20, 28]$, positive term spread, moderate drawdown), blocked in acute crises ($VIX > 30$, funding > p90, HY-IG > 50bps)—defensive carry, not a crisis refuge (Diep et al., 2021). All z-score parameters estimated on the development set only. Routing priority: USD→Gold→MBS→Cash (default). Thresholds optimized by Calmar-ratio grid search on CV folds.

Backtest Results (1.5× Leverage)

Table 2: Strategy vs. benchmarks (2019–2021). TC: equity 5bps, cash 2bps, gold 8bps, MBS 20bps.

	EWS Strategy	60/40	Buy & Hold
CAGR	35.2%	18.1%	24.2%
Sharpe	1.93	1.44	1.38
Max Drawdown	−12.1%	−21.5%	−27.7%
Calmar	2.92	0.84	0.87
Sortino	2.10	—	—
Turnover	3.9/yr	—	—

Allocation: 103 weeks levered equity (85.8%), 8 gold (6.7%), 8 cash (6.7%), 1 MBS (0.8%). During the **COVID crash** (Feb–Apr 2020), USD triggers fired first (funding stress, DXY rally, VRP spike), routing to cash and gold; MBS correctly blocked ($VIX > 30$). In the **recovery** (Apr–Dec 2020), allocations returned to 1.5× equity, capturing the V-shaped rebound. During **reflation** (early 2021), full equity exposure was maintained without false alarms. The core thesis is validated: leverage amplifies returns in calm periods (103 weeks), while the EWS limits drawdown to −12.1% versus −27.7% for unhedged buy-and-hold.

Limitations

Test holdout spans only 120 weeks with 9 regime switches—results are not yet statistically robust. Folds 3–5 have 2–10 positives; n_{pos} -weighting mitigates but does not eliminate CV noise. Proxy constraints (MXUS≠MSCI World, log-ratios≠OAS) and weekly frequency limit real-world applicability. The 2022 inflation/rate-shock regime—the strongest out-of-sample test for gold routing—is absent. At 1.5× leverage, the system’s 35% recall gap (missing roughly one-in-three risk-off episodes) is a genuine concern that must be weighed against the superior risk-adjusted performance.